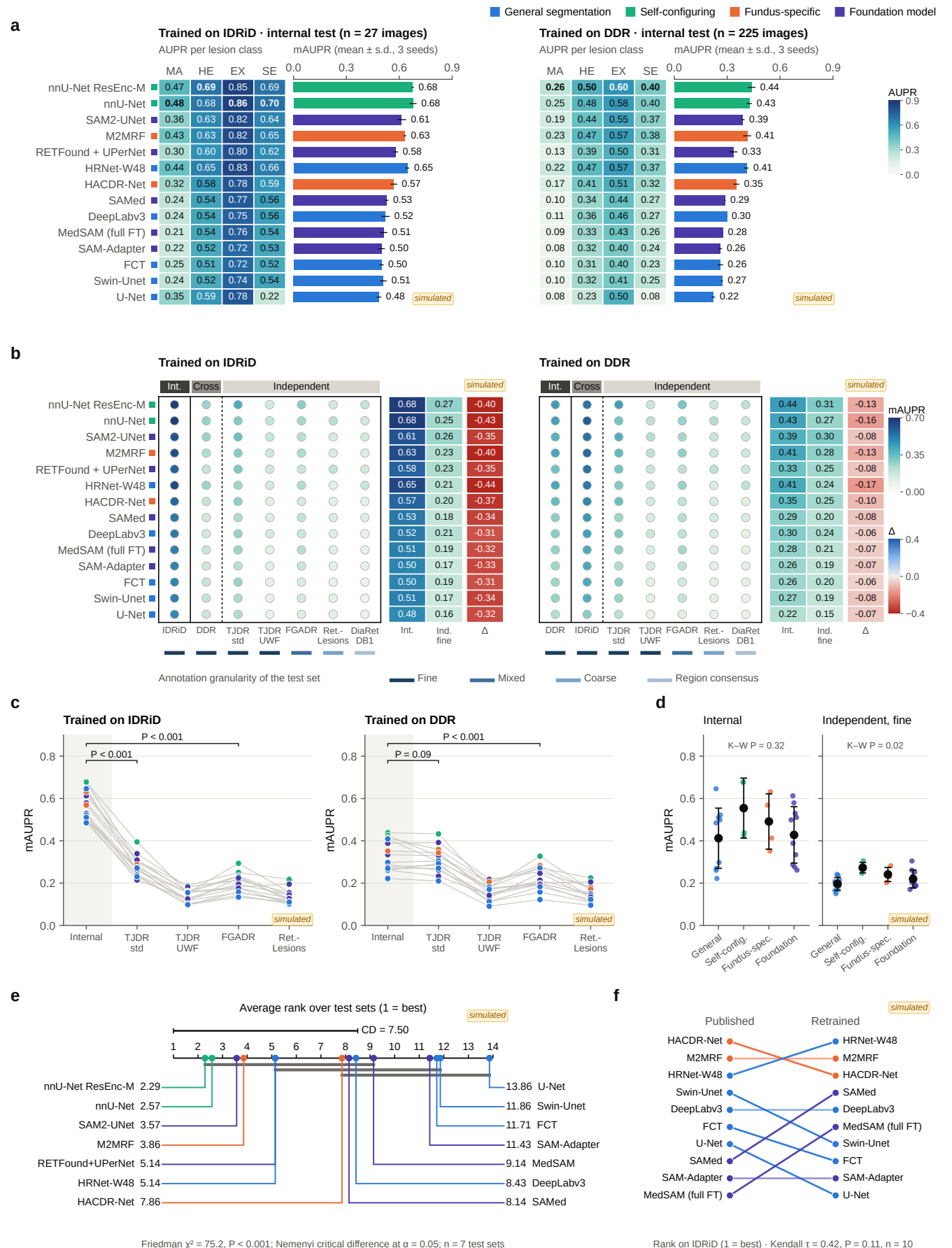

General segmentation
Self-configuring
Fundus-specific
Foundation model
a
Trained on IDRiD · internal test (n = 27 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
nnU-Net ResEnc-M 0.47 0.69 0.85 0.69 0.68
nnU-Net 0.48 0.68 0.86 0.70 0.68
SAM2-UNet 0.36 0.63 0.82 0.64 0.61
M2MRF 0.43 0.63 0.82 0.65 0.63
RETFound + UPerNet 0.30 0.60 0.80 0.62 0.58
HRNet-W48 0.44 0.65 0.83 0.66 0.65
HACDR-Net 0.32 0.58 0.78 0.59 0.57
SAMed 0.24 0.54 0.77 0.56 0.53
DeepLabv3 0.24 0.54 0.75 0.56 0.52
MedSAM (full FT) 0.21 0.54 0.76 0.54 0.51
SAM-Adapter 0.22 0.52 0.72 0.53 0.50
FCT 0.25 0.51 0.72 0.52 0.50
Swin-Unet 0.24 0.52 0.74 0.54 0.51
U-Net 0.35 0.59 0.78 0.22 0.48
simulated
Trained on DDR · internal test (n = 225 images)
AUPR per lesion class
mAUPR (mean ± s.d., 3 seeds)
MA HE EX SE
0.26 0.50 0.60 0.40 0.44
0.25 0.48 0.58 0.40 0.43
0.19 0.44 0.55 0.37 0.39
0.23 0.47 0.57 0.38 0.41
0.13 0.39 0.50 0.31 0.33
0.22 0.47 0.57 0.37 0.41
0.17 0.41 0.51 0.32 0.35
0.10 0.34 0.44 0.27 0.29
0.11 0.36 0.46 0.27 0.30
0.09 0.33 0.43 0.26 0.28
0.08 0.32 0.40 0.24 0.26
0.10 0.31 0.40 0.23 0.26
0.10 0.32 0.41 0.25 0.27
0.08 0.23 0.50 0.08 0.22
simulated
AUPR
b
Trained on IDRiD
Int. Cross Independent
IDRiD DDR TJDR std TJDR UWF FGADR Ret.-Lesions DiaRet DB1
Int. Ind. fine Δ
nnU-Net ResEnc-M 0.68 0.27 -0.40
nnU-Net 0.68 0.25 -0.43
SAM2-UNet 0.61 0.26 -0.35
M2MRF 0.63 0.23 -0.40
RETFound + UPerNet 0.58 0.23 -0.35
HRNet-W48 0.65 0.21 -0.44
HACDR-Net 0.57 0.20 -0.37
SAMed 0.53 0.18 -0.34
DeepLabv3 0.52 0.21 -0.31
MedSAM (full FT) 0.51 0.19 -0.32
SAM-Adapter 0.50 0.17 -0.33
FCT 0.50 0.19 -0.31
Swin-Unet 0.51 0.17 -0.34
U-Net 0.48 0.16 -0.32
simulated
Trained on DDR
Int. Cross Independent
DDR IDRiD TJDR std TJDR UWF FGADR Ret.-Lesions DiaRet DB1
Int. Ind. fine Δ
0.44 0.31 -0.13
0.43 0.27 -0.16
0.39 0.30 -0.08
0.41 0.28 -0.13
0.33 0.25 -0.08
0.41 0.24 -0.17
0.35 0.25 -0.10
0.29 0.20 -0.08
0.30 0.24 -0.06
0.28 0.21 -0.07
0.26 0.19 -0.07
0.26 0.20 -0.06
0.27 0.19 -0.08
0.22 0.15 -0.07
simulated
mAUPR
Δ
Annotation granularity of the test set
Fine
Mixed
Coarse
Region consensus
c
Trained on IDRiD
P < 0.001
P < 0.001
mAUPR
Internal TJDR std TJDR UWF FGADR Ret.-Lesions
simulated
Trained on DDR
P < 0.001
P = 0.09
mAUPR
Internal TJDR std TJDR UWF FGADR Ret.-Lesions
simulated
d
Internal
K–W P = 0.32
Independent, fine
K–W P = 0.02
mAUPR
General Self-config. Fundus-spec. Foundation
simulated
e
Average rank over test sets (1 = best)
simulated
CD = 7.50
1 2 3 4 5 6 7 8 9 10 11 12 13 14
nnU-Net ResEnc-M 2.29
nnU-Net 2.57
SAM2-UNet 3.57
M2MRF 3.86
RETFound+UPerNet 5.14
HRNet-W48 5.14
HACDR-Net 7.86
13.86 U-Net
11.86 Swin-Unet
11.71 FCT
11.43 SAM-Adapter
9.14 MedSAM
8.43 DeepLabv3
8.14 SAMed
Friedman χ² = 75.2, P < 0.001; Nemenyi critical difference at α = 0.05; n = 7 test sets
f
simulated
Published
Retrained
HACDR-Net
M2MRF
HRNet-W48
Swin-Unet
DeepLabv3
FCT
U-Net
SAMed
SAM-Adapter
MedSAM (full FT)
HRNet-W48
M2MRF
HACDR-Net
SAMed
DeepLabv3
MedSAM (full FT)
Swin-Unet
FCT
SAM-Adapter
U-Net
Rank on IDRiD (1 = best) · Kendall τ = 0.42, P = 0.11, n = 10